

Qatar-2b

R-1221

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_210306 · created 2026-07-31T08:49:22+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459279.94561 +/- 0.00099 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.017
Transit depth (Rp/Rs)^2	3.81 +/- 0.65 [%]
Orbital Inclination (inc)	88.09 +/- 1.31
Airmass coefficient 1 (a1)	1.129 +/- 0.022
Airmass coefficient 2 (a2)	-0.092 +/- 0.057
Scatter in the residuals of the lightcurve fit is	1.99 %
Best Comparison Star	#1 - [588, 187]
Optimal Method	PSF photometry
Transit Duration (day)	0.0774 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.195 ± 0.017 vs 0.16327 (deviation 1.86σ)
Duration fit vs published	0.0774 d vs 0.0767 d (deviation 0.3σ)
Mid-time O-C	2.7 min
Depth SNR	11.4
Residual scatter	1.99 %

Light curve

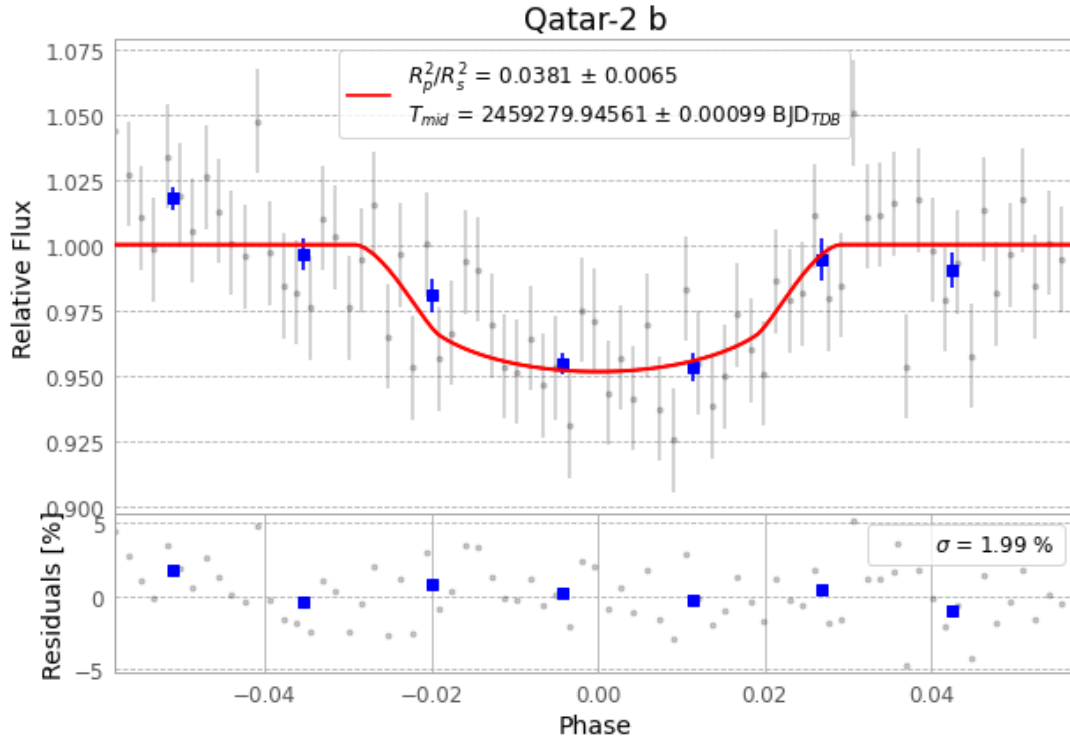


Figure 1 — FinalLightCurve_Qatar-2 b_2021-03-06.png (SHA-256 b418c308c777b04e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [560, 168], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dff6dc20755a58...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), Qatar-2210306084212.FITS → Qatar-2210306122412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-210306.log (b5124f336bc5...), FinalLightCurve_Qatar-2 b_2021-03-06.png (b418c308c777...), FinalLightCurve_Qatar-2 b_2021-03-06.csv (9c2552c66219...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 08:49Z.